LIMIT HEIGHTS OF COMPLETE INTERSECTIONS IN TORIC VARIETIES

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Abstract. Using Deligne pairings of adelic line bundles and equidistribution of small points, we prove Gualdi's conjecture on limit heights of complete intersection cycles in toric varieties. Our result strengthens the original torsion-point formulation to generic sequences of small points.

1. Introduction

In classical algebraic geometry, Bézout's theorem states that generically the intersection of m hypersurfaces of degrees $d_1, \dots, d_m$ in $\mathbb{P}^n$ is of degree $d_1 \dots d_m$. If the hypersurfaces are defined over a number field, it is natural to ask for the Weil height of the intersection. The arithmetic Bézout theorem of Bost, Gillet and Soulé (see [BGS94]) only provides an upper bound for this height. In fact, the height of the intersection is not determined by the heights and degrees of the hypersurfaces. Since Weil heights are also connected to special values of L-functions and periods (see e.g. [PP24, Section 4] or [Mai00; CM00]), it is nevertheless desirable to have formulas for their exact values depending on additional data.

Our starting point is Gualdi's formula in [Gua18b], which computes the height of a hypersurface in a toric variety in terms of the defining Laurent polynomial. However, Gualdi observed in his thesis [Gua18a] that the analogous formula fails to compute the height of the complete intersection of $m \geqslant 2$ hypersurfaces. As a remedy to this issue, he suggested translating the hypersurfaces by torsion points of the torus. He conjectured that the analogous formula would give the limit height of these intersections.

Let us briefly introduce the necessary prerequisites to state his conjecture which is the main result of our paper. Let $K \subseteq \overline{\mathbb{Q}}$ be a number field and write Ω_K for its set of places. For $v \in \Omega_K$, put

$$n_v = \frac{[K_v : \mathbb{Q}_v]}{[K : \mathbb{Q}]}.$$

Let $\mathbb{T} = \mathbb{G}_m^n$, with character and cocharacter lattices

$$M = \mathrm{Hom}(\mathbb{T}, \mathbb{G}_m) \qquad \text{and} \qquad N = \mathrm{Hom}(\mathbb{G}_m, \mathbb{T}),$$

respectively. Thus a Laurent polynomial on $\mathbb{T}$ is an element of the group algebra $K[M]$. A nonzero Laurent polynomial $f \in K[M]$ defines a hypersurface in $\mathbb{T}$. Given a projective toric compactification $\mathbb{T} \subseteq T$, we denote by $V(f)$ the closure in T of its divisor on $\mathbb{T}$, with its natural multiplicities. For $u \in \mathbb{T}(\overline{\mathbb{Q}})$, the notation $uV(f)$ denotes its translate under the torus action on T. Any semipositive toric adelic divisor $\overline{D}$ on T is encoded at each place v by a concave roof function $\theta_{\overline{D},v}$ on the polytope

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associated to its underlying toric divisor. At every place v, the v-adic Ronkin roof function of f is a concave function $\rho_{f,v}^{\vee}\colon \mathrm{NP}(f) \to \mathbb{R}$ on the Newton polytope of f. At a non-archimedean place the function $\rho_{f,v}^{\vee}$ is easily determined by the valuations of the coefficients of f. The collection of Ronkin functions defines a semipositive toric adelic divisor, denoted by $\overline{R(f)}$. Arithmetic intersection numbers of semipositive toric adelic divisor can be computed combinatorially by using mixed integrals. Mixed integrals and further convex geometry tools are discussed in Section 3. Finally, we equip

$$\mathbb{T}^m \subseteq (\mathbb{P}^n)^m$$

with the standard Weil height associated to

$$\overline{P} = \sum_{i=1}^{m} \pi_i^* \overline{H}_\infty,$$

where $\overline{H}_\infty$ is the hyperplane at infinity with its canonical toric metric. A sequence in $\mathbb{T}^m(\overline{\mathbb{Q}})$ is called generic if it eventually avoids every proper closed subvariety, and small if its height with respect to $\overline{P}$ tends to zero.

Theorem A (Theorem 4.6)**.** *Let $1 \leqslant m \leqslant n$, let $f_1, \dots, f_m \in K[M]$ be nonzero Laurent polynomials, and let T be a projective toric variety with dense torus $\mathbb{T} = \mathbb{G}_m^n$. For each i, let*

$$V_i := V(f_i)$$

be the closure in T of the hypersurface defined by f_i. Let $u_j = (u_{1,j}, \dots, u_{m,j}) \in \mathbb{T}^m(\overline{\mathbb{Q}})$ be a generic sequence which is small for the standard Weil height on $\mathbb{T}^m$, and let $\overline{D}_0, \dots, \overline{D}_{n-m}$ be semipositive toric adelic divisors on T. Write $\theta_{k,v}$ for the v-adic roof function of $\overline{D}_k$, and $\rho_{i,v}^{\vee}$ for the v-adic Ronkin roof function of f_i. For all sufficiently large j, the translated hypersurfaces

$$u_{1,j}V_1, \dots, u_{m,j}V_m$$

meet properly. Denote their intersection cycle by $Z_j := (u_{1,j}V_1) \cdots (u_{m,j}V_m) \cdot \mathbb{T}_{\overline{\mathbb{Q}}}$. Then

$$\begin{aligned}\lim_{j\to\infty} \widehat{\deg}\left(\overline{D}_0, \dots, \overline{D}_{n-m} \mid Z_j\right) &= \widehat{\deg}\left(\overline{D}_0, \dots, \overline{D}_{n-m}, \overline{R(f_1)}, \dots, \overline{R(f_m)} \;\middle|\; T\right) \\ &= \sum_{v\in\Omega_K} n_v \,\mathrm{MI}_M\left(\theta_{0,v}, \dots, \theta_{n-m,v}, \rho_{1,v}^{\vee}, \dots, \rho_{m,v}^{\vee}\right).\end{aligned}$$

Here the middle intersection number may be computed on any common projective toric refinement of T on which the Newton polytopes $\mathrm{NP}(f_i)$ define Cartier divisors; by the projection formula, its value is independent of this choice.

Gualdi formulated the conjecture for a generic sequence of torsion points in [Gua18a]. We prove the stronger statement for generic sequences of points whose Weil height tends to zero. The first equality in the theorem says that the typical mixed height of the translated complete-intersection cycle is the arithmetic intersection number obtained by replacing each hypersurface by its Ronkin adelic divisor. The second equality is the toric arithmetic intersection formula, which expresses this number as an adelic sum of mixed integrals. In the course of the proof we use convex geometry methods and in fact first show the equality of the limit to the combinatorial expression.

Gualdi and Sombra have proved partial results in this direction. In [GS25] they prove the above result in the case $T = \mathbb{P}^2$, $m = 2$, and $f_1(x_1, x_2) = f_2(x_1, x_2) =$

$x_1 + x_2 + 1$. Moreover, for this example, they show that both sides of the identity in question are equal to the intriguing value $\frac{2\zeta(3)}{3\zeta(2)}$. They have also solved the $m = 2$ case of the conjecture in [GS26]. Their methods fundamentally differ from ours. In summary, their approach relies on local logarithmic equidistribution as in [DH24] while we rely on adelic line bundles and their Deligne pairings as developed by Yuan and Zhang in [YZ26], and a variant of Yuan's equidistribution theorem in [Yua08].

A prior version of essentially the same proof is contained in [DHS24], where we used the theory of adelic curves and globally valued fields introduced by Chen-Moriwaki [CM20] and Ben Yaacov-Hrushovski [Ben+24] respectively. We have since found it advantageous to split the contents into two separate self-contained papers. The present proof replaces the formalism of continuity of heights over globally valued fields with a direct equidistribution argument involving Deligne pairings.

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2. Adelic line bundles and Deligne pairings

We briefly recall the adelic intersection theory of Yuan and Zhang [YZ26]. By a projective variety over a field K, we mean an integral projective scheme over $\operatorname{Spec} K$. By a quasi-projective variety over K, we mean an open subscheme of a projective variety over K. Tensor products of line bundles are written additively, and a bar indicates adelic metric data. Throughout the section let K be a number field.

2.1. **Adelic line bundles and intersections.** Arithmetic adelic divisors on quasi-projective varieties are defined as certain limits of metrized line bundles on its compactifications. Let U be a quasi-projective variety over K. Then a model of U is an integral, quasi-projective, flat scheme $\mathscr{U}$ over $\operatorname{Spec} \mathcal{O}_K$, together with an isomorphism between U and the generic fibre of $\mathscr{U}$ over $\operatorname{Spec} \mathcal{O}_K$. Fix a projective compactification $\mathscr{U} \subseteq \mathscr{X}_0$, and an effective arithmetic divisor $\overline{E}_0$ whose underlying Cartier divisor is supported in $\mathscr{X}_0 \setminus \mathscr{U}$ and whose Green function is strictly positive. Model arithmetic $\mathbb{Q}$-divisors on $\mathscr{U}$ are obtained by taking the filtered colimit over all projective compactifications of $\mathscr{U}$. We define an extended norm on the space of model arithmetic $\mathbb{Q}$-divisors on $\mathscr{U}$ by

$$\|\overline{D}\|_{\overline{E}_0} := \inf\{\varepsilon \in \mathbb{Q}_{>0} : -\varepsilon\overline{E}_0 \leqslant \overline{D} \leqslant \varepsilon\overline{E}_0\}.$$

Different choices of $\overline{E}_0$ give equivalent norms. The completion of the space of model divisors with respect to this norm is the group of arithmetic adelic divisors on $\mathscr{U}$ [YZ26, §2.4]. Passing to the associated line bundles and then taking the filtered colimit over arithmetic models of U gives the group $\widehat{\mathrm{Pic}}(U)$ of adelic line bundles [YZ26, §2.5].

We define geometric adelic divisors similarly. Given a quasi-projective K-variety U, taking the filtered colimit of divisors over all projective compactifications of U gives the group of model geometric divisors. This colimit carries an extended norm and the elements of the completion are called geometric adelic divisors. An arithmetic adelic divisor defines a geometric adelic divisor by taking the generic fibre. Given an arithmetic adelic line bundle $\overline{L}$ we denote by $\widetilde{L}$ the underlying geometric adelic line bundle. Unless explicitly stated, the term adelic line bundle will refer to arithmetic adelic line bundles.

An adelic divisor $\overline{E}$ is *strongly nef* if it is represented by a Cauchy sequence of nef model divisors. It is *nef* if there exists a strongly nef adelic divisor $\overline{D}$ such that $m\overline{E}+\overline{D}$ is strongly nef for every $m \geqslant 1$, and it is *integrable* if it is the difference of two strongly nef adelic divisors. We apply the same terminology to the line bundles induced by adelic divisors. We write $\widehat{\mathrm{Pic}}(U)_{\mathrm{nef}}$ and $\widehat{\mathrm{Pic}}(U)_{\mathrm{int}}$ for the corresponding subsets of $\widehat{\mathrm{Pic}}(U)$. The pullback of adelic line bundles preserves model, strongly nef, nef, and integrable adelic line bundles [YZ26, §2.5.5].

Proposition 2.1 ([YZ26, Prop. 4.1.1])**.** *If X is a quasi-projective K-variety of dimension n, the intersection pairing on projective models extends to a multilinear pairing*

$$\widehat{\mathrm{Pic}}(X) \times \widehat{\mathrm{Pic}}(X)^n_{\mathrm{int}} \longrightarrow \mathbb{R}, \qquad (\overline{L}_0, \dots, \overline{L}_n) \longmapsto \overline{L}_0 \cdots \overline{L}_n.$$

It is symmetric in the last n entries, and in all entries when $\overline{L}_0$ is integrable. If all factors are nef, then the intersection is nonnegative.

Let X be a projective variety over K, let $\overline{L}_0, \dots, \overline{L}_d$ be integrable adelic line bundles on X, and let Z be a d-cycle on $X_{\overline{\mathbb{Q}}}$. Choose a finite extension K'/K and a cycle $Z_{K'}$ on $X_{K'}$ whose base change to $\overline{K}$ is Z. We define the arithmetic degree of Z by

$$\widehat{\deg}\left(\overline{L}_0, \dots, \overline{L}_d \mid Z\right). := \widehat{\deg}\left(\overline{L}_0, \dots, \overline{L}_d \mid Z_{K'}\right).$$

By the projection formula, this quantity is independent of K'. This specializes to give the notion of the height of an algebraic point. For an adelic line bundle $\overline{L}$ and an algebraic point $x \in X(\overline{\mathbb{Q}})$, we denote its height by

$$h_{\overline{L}}(x) = \widehat{\deg}(\overline{L} \mid \{x\}).$$

If $Z \subseteq X$ is a projective integral subvariety of dimension r and $(\widetilde{L}|_Z)^r \neq 0$, its normalised height is

$$h_{\overline{L}}(Z) = \frac{(\overline{L}|_Z)^{r+1}}{(r+1)(\widetilde{L}|_Z)^r}.$$

These normalizations make the height of a zero-dimensional subvariety agree with the height of its geometric points; see [YZ26, §5.3.1].

2.2. **The Deligne pairing.** Let $f\colon X \to Y$ be a projective flat morphism of relative dimension n between quasi-projective K-varieties, with X integral and Y normal. The classical Deligne pairing extends to a symmetric multilinear functor

$$\langle \cdot, \dots, \cdot \rangle_{X/Y}\colon \widehat{\mathrm{Pic}}(X)^{n+1}_{\mathrm{int}} \longrightarrow \widehat{\mathrm{Pic}}(Y)_{\mathrm{int}}.$$

It preserves strongly nef and nef adelic line bundles. Its underlying line bundle is the usual Deligne pairing, while its adelic metric is obtained as the limit of the Deligne metrics on models. The construction is compatible with normal integral base changes $Y' \to Y$ for which $X \times_Y Y'$ is integral [YZ26, Thm. 4.1.3]. By [YZ26, Lem. 4.2.2], the resulting identity extends by linearity when the fibered product is not integral. More precisely, if X has irreducible components $X_1, \dots, X_r$ with multiplicities $\delta(X_i)$, one may define $\langle \overline{L}_0, \dots, \overline{L}_n \rangle_{X/Y} = \bigotimes_{i=1}^r \langle \overline{L}_0, \dots, \overline{L}_n \rangle_{X_i/Y}^{\otimes \delta(X_i)}$. The properties below remain valid for non-integral X by linearity.

We record the properties used below. Set $\overline{M} = \langle \overline{L}_0, \dots, \overline{L}_n \rangle_{X/Y}$.

(1) *Fibres.* For a $y \in Y(\overline{\mathbb{Q}})$ there is a canonical isometry

$$\overline{M}|_y \simeq \left\langle \overline{L}_0|_{X_y}, \dots, \overline{L}_n|_{X_y} \right\rangle_{X_y/y}.$$

Consequently,

$$h_{\overline{M}}(y) = \widehat{\deg}\left(\overline{L}_0, \dots, \overline{L}_d \mid X_y\right).$$

(2) *Projection formula.* ([YZ26, Lemma 4.6.1 (2)]) If $\overline{N} \in \widehat{\mathrm{Pic}}(Y)_{\mathrm{int}}$, then

$$\left\langle \overline{L}_0, \dots, \overline{L}_{n-1}, f^*\overline{N} \right\rangle_{X/Y} \simeq d\,\overline{N}, \qquad d = (L_0 \cdots L_{n-1})_{X_\eta},$$

with η the generic point of Y.

(3) *Composition.* ([YZ26, Lemma 4.6.1 (1), (3)]) Suppose that $\psi : X \to Y$ and $\pi : Y \to S$ are projective flat morphisms of relative dimensions r and m respectively, with the normality and integrality hypotheses needed to form the pairings. Then,

$$\left\langle \overline{L}_0, \dots, \overline{L}_r, \psi^*\overline{M}_1, \dots, \psi^*\overline{M}_m \right\rangle_{X/S} \simeq \left\langle \langle \overline{L}_0, \dots, \overline{L}_r \rangle_{X/Y}, \overline{M}_1, \dots, \overline{M}_m \right\rangle_{Y/S}.$$

In particular, if η is the generic point of Y and $d = (L_1 \cdots L_r)_{X_\eta}$, then

$$\left\langle \overline{L}_1, \dots, \overline{L}_r, \psi^*\overline{M}_0, \dots, \psi^*\overline{M}_m \right\rangle_{X/S} \simeq d\,\langle \overline{M}_0, \dots, \overline{M}_m \rangle_{Y/S}.$$

(4) *Absolute intersections.* When $Y = \operatorname{Spec} K$, the arithmetic degree of the Deligne pairing recovers the absolute intersection number:

$$\widehat{\deg}\langle \overline{L}_0, \dots, \overline{L}_n \rangle_{X/K} = \overline{L}_0 \cdots \overline{L}_n.$$

2.3. **Equidistribution.** Classical equidistribution theorems are stated as weak convergence of measures. This does not allow us to apply them to functions with logarithmic singularities. In fact, the corresponding logarithmic equidistribution statements may fail. However, variational proofs of equidistribution, such as Yuan's [Yua08], yield an adelic statement reminiscent of logarithmic equidistribution which we need for our proof of Gualdi's conjecture. The following theorem follows from the proof of [Yua26, Thm. 5.4.3], after [Yua08], [Cha21, Lemma 8.2].

Theorem 2.2 (Equidistribution)**.** *Let X be a quasi-projective variety over K and $\overline{L}$ a nef adelic line bundle on X with $\deg_{\widetilde{L}}(X) > 0$ and $\overline{L}^{\dim X+1} = 0$. Let $(x_m)_m$ be a generic sequence in $X(\overline{K})$ with $h_{\overline{L}}(x_m) \to 0$. Then for every adelic line bundle $\overline{M}$ on X we have*

$$\lim_{m\to\infty} h_{\overline{M}}(x_m) = \frac{\overline{M} \cdot \overline{L}^{\dim X}}{\deg_{\widetilde{L}}(X)}.$$

In our application $\overline{L}$ will be the nef and big canonically metrized line bundle $\sum_{i=1}^{N} \pi_i^* \overline{\mathcal{O}(1)}$ on $(\mathbb{P}^n)^m$. It would therefore suffice to apply Yuan's equidistribution result noting that it holds for all adelic line bundles $\overline{M}$ over a quasi-projective open subset.

3. Convex geometry and toric varieties

This section contains basic notions on Ronkin metrics on toric varieties and some calculations in convex geometry. For more details on Ronkin metrics we suggest [Gua18b]. We refer to [BPS14] for both an introduction to toric varieties and an in-depth treatment of the Arakelov geometry of toric varieties.

3.1. **Toric divisors.** Let $\mathbb{T} \simeq \mathbb{G}_m^n$ be a split torus over a number field K. Denote by N its cocharacter lattice and by M its character lattice. Let X_Σ denote the proper toric variety associated to a complete fan Σ in $N_\mathbb{R}$ and let N_Σ denote the compactification of $N_\mathbb{R}$ associated to Σ. Let $\mathbb{T} \simeq X_\Sigma^\circ \subset X_\Sigma$ denote the open $\mathbb{T}$-orbit in X_Σ.

Definition 3.1. A *toric Cartier divisor* on a toric variety X_Σ is defined to be a Cartier divisor which is invariant under the action of the torus $\mu : \mathbb{T} \times X_\Sigma \to X_\Sigma$. This means that a Cartier divisor D is toric if $\mu^* D = \pi^* D$, where $\pi : \mathbb{T} \times X_\Sigma \to X_\Sigma$ denotes the projection map.

Toric Cartier divisors are in bijection with virtual polytopes as explained in Section 3.3 of [BPS14].

Definition 3.2. A *virtual support function* or *virtual polytope* with respect to a complete fan Σ on $N_\mathbb{R}$ is a function $N_\mathbb{R} \to \mathbb{R}$ that is linear and integral on each cone in Σ.

Let v be a place of K. A v-adic Green function g_v for a toric Cartier divisor D is called toric if its restriction to $(X_\Sigma^\circ)_v^{\mathrm{an}}$ factors through the valuation map $\mathrm{trop} : (X_\Sigma^\circ)_v^{\mathrm{an}} \to N_\mathbb{R}$ sometimes also called the tropicalization map. The dependence of trop on $v \in \Omega_K$ is implicit in this notation. Proposition 4.3.10 and Theorem 4.8.1.(1) in [BPS14] imply the following.

Theorem 3.3. *Let D be the toric divisor associated to the virtual support function Ψ. Then the space of toric v-adic Green functions for D is in bijection with continuous functions $\psi : N_\mathbb{R} \to \mathbb{R}$ such that $\psi - \Psi$ is bounded and extends continuously to the compactification N_Σ of $N_\mathbb{R}$. If ψ is concave, then it is enough to require that $\psi - \Psi$ is bounded. Concave functions correspond to semipositive metrics under this bijection.*

If $\overline{D}_v$ is a toric Cartier divisor with a semipositive toric v-adic Green function, Legendre-Fenchel duality associates to ψ a concave function

$$\theta_{\overline{D}_v} : x \mapsto \psi^\vee(x) = \inf_{u \in N_\mathbb{R}} \left(\langle x, u \rangle - \psi(u) \right)$$

on the polytope $\Delta_\Psi \subseteq M_\mathbb{R}$ defined as

$$\Delta_\Psi := \{ x \in M_\mathbb{R} : \langle x, u \rangle \geqslant \Psi(u) \text{ for all } u \in N_\mathbb{R} \}.$$

The function $\theta_{\overline{D}_v} : \Delta_\Psi \to \mathbb{R}$ is called the *roof function* of $\overline{D}_v$.

A collection of continuous functions $(\psi_v)_{v \in \Omega_K}$ defines an adelic metric on the toric divisor corresponding to Ψ if for all v the difference $\psi_v - \Psi$ is bounded and extends continuously to N_Σ, and for almost all v we have $\psi_v = \Psi$. We call an arithmetic divisor $\overline{D}$ on X_Σ a *toric adelic divisor* if the underlying Cartier divisor D is a toric

Cartier divisor, and the metrics on $\overline{D}$ come from a collection of functions $(\psi_v)_{v\in\Omega_K}$ satisfying the above condition.

3.2. **Mixed volumes and mixed integrals.** The vector space $M_\mathbb{R}$ carries a Haar measure normalised in such a way that M has covolume 1. We associate to a compact convex set Δ its volume $\mathrm{vol}_M(\Delta)$ with respect to this Haar measure. Recall that the Minkowski sum of subsets S_1, S_2 of a vector space V is defined by

$$S_1 + S_2 = \{s_1 + s_2 \mid s_1 \in S_1, s_2 \in S_2\}.$$

The volume is a homogeneous polynomial on the Grothendieck group of the space of compact convex sets with Minkowski addition. It can therefore be polarised, cf. [BPS14, Definition 2.7.14].

Definition 3.4. The mixed volume is an n-linear symmetric form on compact convex sets $\Delta_1, \dots, \Delta_n \subseteq M_\mathbb{R}$ (recall that $n = \dim_\mathbb{R} M_\mathbb{R}$) defined by

$$\mathrm{MV}_M(\Delta_1, \dots, \Delta_n) = \sum_{j=1}^{n} (-1)^{n-j} \sum_{1\leqslant i_1<\dots<i_j\leqslant n} \mathrm{vol}_M\left(\Delta_{i_1} + \dots + \Delta_{i_j}\right).$$

It satisfies $\mathrm{MV}_M(\Delta, \dots, \Delta) = n!\,\mathrm{vol}_M(\Delta)$.

Given concave functions θ_1 and θ_2 on the compact convex sets Δ_1 and Δ_2, we define their sup-convolution by

$$(\theta_1 \boxplus \theta_2)(m) = \sup_{m_1+m_2=m} \theta_1(m_1) + \theta_2(m_2).$$

It defines a concave function on $\Delta_1+\Delta_2$. It is defined precisely so that the hypograph of $\theta_1 \boxplus \theta_2$ is the Minkowski sum of the hypographs of θ_1 and θ_2.

Definition 3.5. The mixed integral is a multilinear form on the cone of compactly supported concave functions with sup-convolution and positive scalar multiplication. It maps concave functions $\theta_0, \dots, \theta_n$ on compact convex sets $\Delta_0, \dots, \Delta_n \subseteq M_\mathbb{R}$ to

$$\mathrm{MI}_M(\theta_0, \dots, \theta_n) = \sum_{j=0}^{n} (-1)^{n-j} \sum_{0\leqslant i_0<\dots<i_j\leqslant n} \int_{\Delta_{i_0}+\dots+\Delta_{i_j}} \theta_{i_0} \boxplus \dots \boxplus \theta_{i_j}\,\mathrm{dvol}.$$

It satisfies $\mathrm{MI}_M(\theta, \dots, \theta) = (n+1)! \int_\Delta \theta \; d\,\mathrm{vol}_M$.

If a lattice M is fixed, we omit it from the notation. These notions are helpful to express arithmetic intersection numbers combinatorially.

Theorem 3.6 ([BPS14, Theorem 5.2.5])**.** *Let $\overline{D}_0, \dots, \overline{D}_n$ be semipositive toric adelic divisors on X_Σ. Then,*

$$\widehat{\deg}\left(\overline{D}_0, \dots, \overline{D}_n \mid X_\Sigma\right) = \sum_{v\in\Omega_K} n_v\, \mathrm{MI}_M\left(\theta_{0,v}, \dots, \theta_{n,v}\right).$$

For a nonzero Laurent polynomial $f \in K[M]$, we denote by $V(f)$ the Zariski closure in X_Σ of the vanishing locus of f on X_σ°. Let $\mathrm{NP}(f)$ be the Newton polytope of $f = \sum_m c_m \chi^m$, i.e., the convex hull of the exponents m with $c_m \neq 0$. Then, $\mathrm{NP}(f)$ defines a Cartier divisor on a suitable toric modification of X_Σ° such that f gives rise to a regular section (see [BPS14, Section 3.4] and [Gua18b, Lemma 5.1]). This section vanishes precisely on $V(f)$ by [Gua18b, Theorem 4.3].

We now define the Ronkin metric on the divisor $D_{\mathrm{NP}(f)}$ corresponding to $\mathrm{NP}(f)$. For this we use that the fibres $\mathrm{trop}^{-1}(u)$ of the tropicalization map $\mathrm{trop} : \mathbb{T}_v^{\mathrm{an}} \to N_\mathbb{R}$ carry a natural choice of probability measure σ_u. In the non-Archimedean case, it is

concentrated on the Gauss point over u. In the Archimedean case, it is induced by the Haar measure on $(S^1)^n$.

Definition 3.7. [Gua18b, Definition 2.7] Let f be a nonzero Laurent polynomial over K. The Ronkin function of f over a place v is the map $\rho_{f,v} : N_{\mathbb{R}} \to \mathbb{R}$ defined by

$$\rho_{f,v} : u \mapsto \int_{\mathrm{trop}^{-1}(u)} -\log|f(x)|_v \, \mathrm{d}\sigma_u(x).$$

By [Gua18b, Proposition 2.10], $\rho_{f,v}$ is a concave continuous function with bounded difference from $\Psi_{\mathrm{NP}(f)}$. By Theorem 3.3, it defines a semipositive toric Green function for $D_{\mathrm{NP}(f)}$. Thus, the collection of v-adic Ronkin functions, for $v \in \Omega_K$, gives rise to a semipositive toric adelic divisor that we denote by $\overline{R(f)}$. For the proof see Lemma 5.11 in [Gua18b]. We call $\overline{R(f)}$ the *Ronkin divisor associated to* f. We denote the underlying geometric divisor by $R(f)$ and also refer to it as the (geometric) Ronkin divisor of f.

Theorem 3.8 (variation of [Gua18b, Theorem 5.12])**.** *Let T be a projective toric variety. Let $\overline{D}_0, \dots, \overline{D}_{n-1}$ be semipositive toric adelic divisors on T. Let f be a Laurent polynomial and let $V(f)$ be the closure in T of the hypersurface in $\mathbb{T}$ defined by f with appropriate multiplicities. Then,*

$$\widehat{\deg}\left(\overline{D}_0, \dots, \overline{D}_{n-1} \mid V(f)\right) = \widehat{\deg}\left(\overline{D}_0, \dots, \overline{D}_{n-1}, \overline{R(f)} \mid T\right).$$

3.3. **Functoriality.**

Definition 3.9. Let $\gamma : V \to W$ be a homomorphism of finite dimensional real vector spaces and $f : \Delta \to \mathbb{R}$ be a continuous concave function with compact domain $\Delta \subseteq V$. We define the direct image of f along γ by

$$\gamma_* f(w) = \max_{v \in \gamma^{-1}(w)} f(v).$$

It is a continuous concave function $\gamma_* f : \gamma(\Delta) \to \mathbb{R}$.

Lemma 3.10. *Let $\gamma : M \to M'$ be a map of lattices and let $K[\gamma] : K[M] \to K[M']$ be the induced map on the rings of Laurent polynomials. Then for any $f \in K[M] \setminus \{0\}$ such that $K[\gamma](f) \neq 0$, we have*

$$\rho_{K[\gamma](f)} = \rho_f \circ \gamma^\vee$$

and

$$\rho^\vee_{K[\gamma](f)} = \gamma_* \rho^\vee_f.$$

Proof. Let $u \in N'_{\mathbb{R}}$. Then, the torus morphism induced by γ maps the fibre $\mathrm{trop}^{-1}(u)$ to $\mathrm{trop}^{-1}(\gamma^\vee(u))$, and pushes forward the natural measure σ_u to $\sigma_{\gamma^\vee(u)}$. Thus, the first statement follows from the definition of $K[\gamma](f)$. The second statement follows from the first using a dualised version of [BPS14, Proposition 2.3.8(1)]. □

We now prove generalizations of Lemma 1.11 and Proposition 1.12 in [Gua18b]. We apply them to write the formulas from the preceding section in the form used in [Gua18a, Conjecture 6.4.4].

Lemma 3.11. *Let $L \subseteq M$ be a saturated k-dimensional sublattice and let $P = M/L$ be the quotient lattice. Assume that $\Delta_1, \dots, \Delta_k \subseteq L_{\mathbb{R}}$ are polytopes and let $\pi : M_{\mathbb{R}} \to P_{\mathbb{R}}$ be the quotient map. Let $A_1, \dots, A_{n-k}$ be polytopes in $M_{\mathbb{R}}$. Then,*

$$\mathrm{MV}_M(\Delta_1, \dots, \Delta_k, A_1, \dots, A_{n-k}) = \mathrm{MV}_L(\Delta_1, \dots, \Delta_k) \cdot \mathrm{MV}_P(\pi(A_1), \dots, \pi(A_{n-k})).$$

Proof. By the definition of mixed volume one obtains

$$\begin{aligned}\mathrm{MV}_M\left(\Delta_1,\dots,\Delta_k,A_1,\dots,A_{n-k}\right)\\ =\sum_{J\subseteq\{1,\dots,n-k\}}(-1)^{n-|J|}\sum_{I\subseteq\{1,\dots,k\}}(-1)^{|I|}\operatorname{vol}_M\left(\Delta_I+A_J\right).\end{aligned}$$

Here Δ_I and A_J are taken to denote respectively $\sum_{i\in I}\Delta_i$ and $\sum_{j\in J}A_j$, with the convention $\Delta_\emptyset=A_\emptyset=\{0\}$. It now suffices to show that for all convex bodies A in $M_\mathbb{R}$ we have

$$\sum_{I\subseteq\{1,\dots,k\}}(-1)^{|I|}\operatorname{vol}_M\left(\Delta_I+A\right)=(-1)^k\,\mathrm{MV}_L\left(\Delta_1,\dots,\Delta_k\right)\operatorname{vol}_P(\pi(A)).$$

For this take any $p\in P_\mathbb{R}$ and denote by A_p the preimage of p in $M_\mathbb{R}$ intersected with A. Note that under the identification $M_\mathbb{R}\simeq L_\mathbb{R}\times P_\mathbb{R}$, A_p can be viewed as a subset of $L_\mathbb{R}$, and since $\operatorname{vol}_M=\operatorname{vol}_L\times\operatorname{vol}_P$, by Fubini's theorem and the fact that $\Delta_I\subseteq L_\mathbb{R}$ we get

$$\operatorname{vol}_M(\Delta_I+A)=\int_{P_\mathbb{R}}\operatorname{vol}_L(\Delta_I+A_p)\,\mathrm{dvol}_P(p).$$

Thus we may write $\sum_{I\subseteq\{1,\dots,k\}}(-1)^{|I|}\operatorname{vol}_M(\Delta_I+A)$ as

$$\int_{\pi(A)}\sum_{I\subseteq\{1,\dots,k\}}(-1)^{|I|}\operatorname{vol}_L(\Delta_I+A_p)\,\mathrm{dvol}_P(p).$$

In order to conclude, we need to show that

$$\sum_{I\subseteq\{1,\dots,k\}}(-1)^{|I|}\operatorname{vol}_L(\Delta_I+R)=(-1)^k\,\mathrm{MV}_L\left(\Delta_1,\dots,\Delta_k\right)$$

for any polytope R in $L_\mathbb{R}$. For this we decompose the expression by writing out the volumes as mixed volumes, decomposing each of them, and ordering the terms by the number of each Δ_i occurring in the expansion.

$$\begin{aligned}\sum_{I\subseteq\{1,\dots,k\}}(-1)^{|I|}\operatorname{vol}_L(\Delta_I+R)\\ =\frac{1}{k!}\sum_{I\subseteq\{1,\dots,k\}}(-1)^{|I|}\sum_{\substack{a\in S_k\\ \mathrm{Supp}(a)\subset I}}\binom{k}{a_1,\dots,a_k,k-|a|}\mathrm{MV}_L(\Delta^a,R^{k-|a|}).\end{aligned}$$

Here, S_k denotes the set of nonnegative integer sequences $a=(a_1,\dots,a_k)$ with $a_1+\ldots+a_k\leqslant k$. For such a sequence, we define $|a|=a_1+\ldots+a_k$, $\mathrm{Supp}(a)=\{i\in\{1,\dots,k\}\mid a_i\neq 0\}$, and we write $\mathrm{MV}_L(\Delta^a,R^{k-|a|})$ for the mixed volume where each Δ_i appears a_i times, and R appears $k-|a|$ times. The multinomial coefficient

$$\binom{k}{a_1,\dots,a_k,k-|a|}=\frac{k!}{a_1!\cdot\ldots\cdot a_k!\cdot(k-|a|)!}$$

is the number of partitions of a set of k elements into $k+1$ distinguished sets of cardinality $a_1,\dots,a_k,k-|a|$. Therefore,

$$\sum_{I\subseteq\{1,\dots,k\}}(-1)^{|I|}\operatorname{vol}_L(\Delta_I+R)=\sum_{a\in S_k}\frac{\mathrm{MV}_L(\Delta^a,R^{k-|a|})}{a_1!\cdot\ldots\cdot a_k!\cdot(k-|a|)!}\sum_{I\supseteq\mathrm{Supp}(a)}(-1)^{|I|},$$

where for a fixed sequence $a\in S_k$, the sum $\sum_{I\supseteq\mathrm{Supp}(a)}(-1)^{|I|}$ is nonzero if and only if $\mathrm{Supp}(a)=\{1,\dots,k\}$, *i.e.* if $a=(1,\dots,1)$. This concludes the proof. □

If $\Delta \subseteq M_{\mathbb{R}}$ is a polytope, we denote by 0_Δ the function $0_\Delta : \Delta \to \mathbb{R}$ that is identically zero.

Lemma 3.12. *With the same hypotheses as in Lemma 3.11, let g_i be concave functions on each A_i. Then,*

$$\mathrm{MI}_M\left(0_{\Delta_1}, \dots, 0_{\Delta_k}, g_1, \dots, g_{n-k+1}\right) = \mathrm{MV}_L\left(\Delta_1, \dots, \Delta_k\right) \cdot \mathrm{MI}_P\left(\pi_* g_1, \dots, \pi_* g_{n-k+1}\right).$$

Proof. The reduction to the Lemma 3.11 is precisely as in [Gua18b, Proposition 1.12], using [PS08, Proposition IV.5(d)] to reduce mixed integrals to mixed volumes. □

4. Proof of the main theorem

In this section we prove Gualdi's conjecture. The general strategy consists of studying the map sending $(x_1, \dots, x_m) \in \mathbb{T}^m(\overline{\mathbb{Q}})$ to the height of $x_1 V_1 \cap \cdots \cap x_m V_m$ and observing that for a dense open subset of $\mathbb{T}^m$ this is given by the height with respect to an adelic line bundle. To study its behaviour for a small generic sequence of $(x_1, \dots, x_m) \in \mathbb{T}^m(\overline{\mathbb{Q}})$ we apply an equidistribution theorem. This provides a formula for the limit height in question. We prove the vanishing of certain integrals and convex geometry formulae in order to deduce the equality to the quantity conjectured by Gualdi.

We freely replace toric compactifications by common toric refinements and use the same notation for the pullbacks of divisors and cycles. The projection formula shows that the intersection numbers below are unchanged. Similarly, since intersection numbers of adelic line bundles do not change when passing to a dense open subset we omit restriction to a dense open subset in our notation.

4.1. **Setup.** Let $1 \leqslant m \leqslant n$. Let $f_1, \dots, f_m \neq 0$ be non-invertible Laurent polynomials in n variables with coefficients in a number field $K \subset \overline{\mathbb{Q}}$. The condition of being non-invertible excludes precisely scalar multiples of monomials. Let T be a projective toric variety containing the dense open torus $\mathbb{T} = \mathbb{G}_m^n \subseteq T$ whose defining fan is compatible with the Newton polytopes $\mathrm{NP}(f_i)$ for each i. Fix an ample toric Cartier divisor D on T. Each of the Laurent polynomials f_i defines an effective cycle $V_i \subseteq T$ by extending the cycle it defines on $\mathbb{T}$. We study the height of cycles of the form $x_1 V_1 \cap \ldots \cap x_m V_m$ by viewing them as an algebraically parametrized family over $\mathbb{T}^m$. For $i = 1, \dots, m$, let

$$\pi_i : (\mathbb{P}^n)^m \longrightarrow \mathbb{P}^n$$

be the projection onto the i-th factor. Let $\overline{H}_\infty$ denote the hyperplane at infinity in $\mathbb{P}^n$, equipped with its canonical toric metric, and set

$$\overline{P} := \sum_{i=1}^{m} \pi_i^* \overline{H}_\infty.$$

Denote the coordinates of the i-th factor of $\mathbb{T}^m$ by $w_{i,1}, \dots, w_{i,n}$ and let $z_1, \dots, z_n$ be coordinates of $\mathbb{T}$. Let $g_i = f_i(z_1 w_{i,1}^{-1}, \dots, z_n w_{i,n}^{-1})$. Let X be a toric compactification of $\mathbb{T} \times \mathbb{T}^m$ that dominates $T \times (\mathbb{P}^n)^m$ and such that $\mathrm{NP}(g_i)$ defines Cartier divisors on X for each i. This allows us to view $g_i s_i$ as a section of the line bundle $\mathcal{O}(R(g_i))$. Let $\mathcal{V}_i = \mathrm{div}(g_i s_i)$ be the associated divisor. It is defined such that the fibre of $\mathcal{V}_i$ over $(x_1, \dots, x_n) \in \mathbb{T}^m(\overline{\mathbb{Q}})$ is $x_i V_i$. We let p denote the projection $X \to (\mathbb{P}^n)^m$ and q the projection $X \to T$. For later use in proofs by induction we introduce the cycle $\mathcal{Y}_r = \mathcal{V}_1 \cap \cdots \cap \mathcal{V}_r$.

We first prove that generically the translates of the V_i intersect transversally by proving that the $\mathcal{V}_i$ intersect transversally. A similar result can be found in [GS26, Theorem 1.5].

Lemma 4.1. *The Cartier divisors $\mathcal{V}_1, \dots, \mathcal{V}_m$ form a regular sequence on X. Moreover, their intersection cycle is the closure in X of its restriction to $\mathbb{T} \times \mathbb{T}^m$.*

Proof. Set

$$G := \mathbb{T} \times \mathbb{T}^m \qquad \text{and} \qquad \partial X := X \setminus G.$$

We prove the result by induction on m that $\mathcal{V}_1, \dots, \mathcal{V}_m$ form a regular sequence and that no irreducible component of $\mathcal{Y}_m = \mathcal{V}_1 \cap \ldots \cap \mathcal{V}_m$ is contained in ∂X. For $m = 1$, regularity is immediate, while the closure assertion follows from [Gua18b, Theorem 4.3]. Suppose now that the assertions hold for $m-1$. Since $g_1, \dots, g_{m-1}$ do not involve the coordinates of the m-th parameter torus, there is a closed subscheme W of the product of the remaining torus factors such that

$$Y_{m-1} \cap G = W \times \mathbb{T}.$$

No irreducible component of this product is contained in $\mathcal{V}_m$. Otherwise, at the generic point of a component of W, the Laurent polynomial $f_m(zw^{-1})$ would vanish for every $w \in \mathbb{T}$, contradicting $f_m \neq 0$. Thus $\mathcal{V}_1, \dots, \mathcal{V}_m$ form a regular sequence.

It remains to exclude components on the boundary. Consider the subtorus

$$H := \big\{ (a; \underbrace{a, \dots, a}_{m-1}, b) \mid a, b \in \mathbb{T} \big\} \subseteq G,$$

where the first coordinate acts on the T-factor and the following coordinates act on the parameter factors. The functions $g_1, \dots, g_{m-1}$ are invariant under H. Hence H stabilizes both Y_{m-1} and

$$B_{r-1} := Y_{r-1} \cap \partial X.$$

Since H is connected, it stabilizes every irreducible component of B_{m-1}. On the other hand,

$$g_r\big((a; a, \dots, a, b) \cdot (z, w_1, \dots, w_m)\big) = f_r\big(ab^{-1} z w_m^{-1}\big),$$

and ab^{-1} ranges over all of $\mathbb{T}$. Thus the H-translates of $\mathcal{V}_m$ give the full family of twists of g_m. By [GS26, Lemma 1.4], a generic such twist avoids the generic points of all irreducible components of B_{m-1}. Consequently, no component of B_{m-1} can be contained in $\mathcal{V}_r$. Since Y_m is pure of codimension m, any irreducible component of Y_m contained in ∂X would be an irreducible component of B_{m-1} contained in $\mathcal{V}_m$. Hence no such component exists, and the cycle of Y_m is the closure of its restriction to G. The induction is complete. □

We define $\mathcal{V}$ to be the intersection cycle of the $\mathcal{V}_i$ or in other words

$$\operatorname{div}(s_1 g_1) \cdots \operatorname{div}(s_m g_m) \cdot X.$$

None of its components is contained in the boundary of X. By generic flatness, choose a dense open subset $U \subseteq \mathbb{T}^m$ over which the base change $\mathcal{V}_U \to U$ is flat. Since the fibre of each $\mathcal{V}_i$ is a divisor that is linearly equivalent to the Ronkin divisor $R(f_i)$ every nonempty fibre of $\mathcal{V}_U \to U$ has pure dimension $n - m$, and its degree with respect to a divisor D is

$$\deg_D(\mathcal{V}_u) = R(f_1) \cdots R(f_m) \cdot D^{n-m}.$$

For a point $u \in U$ the fibres of $\mathcal{V}$ and $\mathcal{V}_i$ are related by $\mathcal{V}_u = \bigcap_i (\mathcal{V}_i)_u = \bigcap_i (u_i V_i)$ where the right hand side is the scheme theoretic intersection.

4.2. **The Deligne pairing on the parameter space.** Fix integrable adelic divisors $\overline{D}_0, \dots, \overline{D}_{n-m}$ on T. We will later specialize to toric divisors. For $u = (u_1, \dots, u_m) \in U(\overline{\mathbb{Q}})$, set

$$Z_u := (u_1 V_1) \cdots (u_m V_m) \cdot T_{\overline{\mathbb{Q}}}.$$

We are interested in the limit

$$\lim_{j\to\infty} \widehat{\deg}\left(\overline{D}_0, \dots, \overline{D}_{n-m} \mid Z_{u_j}\right),$$

whose terms are well defined for sufficiently large j. Let $\overline{L}_i = q^*\mathcal{O}(\overline{D}_i)$ and put

$$\overline{M} = \left\langle \overline{L}_0, \dots, \overline{L}_{n-m} \right\rangle_{\mathcal{V}_U/U}.$$

For every point $u \in U(\overline{\mathbb{Q}})$, the fibre property of the Deligne pairing gives

$$\widehat{\deg}\left(\overline{D}_0, \dots, \overline{D}_{n-m} \mid Z_u\right) = h_{\overline{M}}(u).$$

By applying the equidistribution theorem, Theorem 2.2, to the line bundle $\mathcal{O}(\overline{P})$ on U we obtain that

$$\lim_j h_{\overline{M}}(u_j) = \frac{\widehat{\deg}\left(\overline{M}, \mathcal{O}(\overline{P}), \dots, \mathcal{O}(\overline{P}) \mid U\right)}{\deg_{\mathcal{O}(P)}((\mathbb{P}^n)^m)}.$$

We compute

$$c := \deg_{\mathcal{O}(P)}((\mathbb{P}^n)^m) = \left(\sum_{i=1}^m \pi_i^*\mathcal{O}(1)\right)^{nm} = \frac{(mn)!}{(n!)^m}.$$

We obtain a formula for the quantity we are interested in which we summarise as a lemma.

Lemma 4.2. *Under the above assumptions*

$$\begin{aligned}\lim_{j\to\infty} \widehat{\deg}\left(\overline{D}_0, \dots, \overline{D}_{n-m} \mid Z_{u_j}\right) &= c^{-1}\widehat{\deg}\Big(\overline{M}, \underbrace{\mathcal{O}(\overline{P}), \dots, \mathcal{O}(\overline{P})}_{nm} \mid U\Big) \\ &= c^{-1}\widehat{\deg}\Big(\overline{L}_0, \dots, \overline{L}_{n-m}, \underbrace{p^*\mathcal{O}(\overline{P}), \dots, p^*\mathcal{O}(\overline{P})}_{nm} \mid \mathcal{V}_U\Big).\end{aligned}$$

Proof. The first equality is the preceding equidistribution calculation. The second follows from the composition property of Deligne pairings(see 2.2(3)) and the identification of absolute intersections (see 2.2(4)). □

4.3. **Identities between arithmetic intersection numbers.** We relate the above quantities to an arithmetic intersection number on X. This requires the computation of some integrals on Berkovich spaces. Before proceeding to the computation we establish an idenity of measures that serves as a non-Archimedean analogue of Fubini's theorem. Let F be a complete non-Archimedean field, let $\mathbb{T} = \operatorname{Spec} F[M]$ be a split torus, and let S be an F-scheme of finite type. The projection

$$\pi : (S \times \mathbb{T})^{\mathrm{an}} \longrightarrow S^{\mathrm{an}}$$

admits a canonical continuous section

$$\sigma : S^{\mathrm{an}} \longrightarrow (S \times \mathbb{T})^{\mathrm{an}}$$

sending $s \in S$ to the Gauss point over s. If $S = \operatorname{Spec} A$ is affine the map is explicitly defined by sending $s \in S^{\mathrm{an}}$ to $\sigma(s)$ defined by the seminorm

$$\left| \sum_{u \in M} a_u \chi^u \right|_{\sigma(s)} := \max_{u \in M} |a_u|_s, \qquad a_u \in A.$$

The explicit formula immediately implies continuity of the section. Let $\overline{\mathcal{O}(1)}$ be the canonically metrised line bundle on the compactification $\mathbb{P}^n$ of $\mathbb{T}$. Recall that its Monge-Ampère measure is the Dirac measure at the Gauss point. Given a (δ-)form ω of top degree we denote its associated measure by $\mu(\omega)$. Denote the projection from $S \times \mathbb{T}$ to the first and second factor by π_1 and π_2 respectively.

Lemma 4.3. *Let S be a projective variety of dimension d over F. Let $\overline{L}_1, \dots, \overline{L}_d$ be semipositive metrized line bundles on S. Let $\mathbb{T} \simeq \mathbb{G}_m^n$ be a split torus. Then,*

$$\mu\left(c_1(\pi_1^* \overline{L}_1) \dots c_1(\pi_1^* \overline{L}_d) \wedge c_1(\pi_2^* \overline{\mathcal{O}(1)})^n\right) = \sigma_* \mu\left(c_1(\overline{L}_1) \dots c_1(\overline{L}_d)\right).$$

Proof. By approximation we may assume the metrics of $\overline{L}_1, \dots, \overline{L}_d$ to be smooth. Let W be a very affine subset of S, i.e. there exists a closed embedding $W \hookrightarrow \mathbb{G}_m^k$, such that the metric of each $\overline{L}_i$ is defined using the tropicalization map $W^{\mathrm{an}} \to \mathbb{R}^k$. We furthermore have a tropicalization map $(W \times \mathbb{T})^{\mathrm{an}} \to \mathbb{R}^k \times N_{\mathbb{R}}$. We note that the image of this morphism is $W^{\mathrm{trop}} \times N_{\mathbb{R}}$. We may approximate the canonical metric on $\overline{\mathcal{O}(1)}$ by smooth metrics. This allows us to deduce the claim by Fubini at the level of the tropicalizations. □

Lemma 4.4. *Let $\overline{R(g_i)}$ denote the Ronkin divisor of g_i on X. Then*

$$\begin{aligned}\widehat{\deg}\left(\mathcal{O}(\overline{R(g_1)}), \dots, \mathcal{O}(\overline{R(g_m)}), \overline{L}_0, \dots, \overline{L}_{n-m}, p^*\mathcal{O}(\overline{P}), \dots, p^*\mathcal{O}(\overline{P}) \mid X\right) \\ = \widehat{\deg}\left(\overline{L}_0, \dots, \overline{L}_{n-m}, p^*\mathcal{O}(\overline{P}), \dots, p^*\mathcal{O}(\overline{P}) \mid \overline{V}\right),\end{aligned}$$

where $\overline{P}$ appears mn times on both sides.

Proof. Let s_i denote the distinguished section of $\mathcal{O}(\overline{R(g_i)})$. The sections $g_i s_i$ of $\mathcal{O}(\overline{R(g_i)})$ have common vanishing locus $\mathcal{V}$. By the iterative definition of the height, the equality of intersection numbers is equivalent to the vanishing of the integrals occurring in the height computations for each place v. Fix a place v and suppress v and the superscript "an" from the notation for the remainder of the proof. These local integrals are of the form

$$\int_{\mathcal{V}_1 \cap \dots \cap \mathcal{V}_{r-1}} \log|g_r s_r| \, c_1\big(\mathcal{O}(\overline{R(g_{r+1})})\big) \dots c_1\big(\mathcal{O}(\overline{R(g_m)})\big) c_1\big(\overline{L}_0\big) \dots c_1\big(\overline{L}_{n-m}\big) \prod_{i=1}^{m} \pi_i^* c_1\big(\overline{\mathcal{O}(1)}\big)^n.$$

Let us write ω for

$$c_1(\mathcal{O}(\overline{R(g_{r+1})})) \dots c_1(\mathcal{O}(\overline{R(g_m)})) c_1(\overline{L}_0) \dots c_1(\overline{L}_{n-m}) \prod_{i \neq r} \pi_i^* c_1(\overline{\mathcal{O}(1)})^n.$$

Let π_r denote the projection from $\mathbb{T} \times \prod_{i=1}^m \mathbb{T}$ to the r-th component $\mathbb{T}_r$ in the second factor. Let $Y_r = \mathcal{V}_1 \cap \dots \cap \mathcal{V}_{r-1} \cap (\mathbb{T} \times \prod_{i=1}^{r-1} \mathbb{T})$. To each of the above integrals we can apply Fubini in the Archimedean case and Lemma 4.3 in the non-Archimedean

case to obtain:

$$\begin{aligned}\int_{\mathcal{V}_1\cap\cdots\cap\mathcal{V}_{r-1}\cap X}\log\|g_rs_r\|\pi_r^*c_1(\overline{\mathcal{O}(1)})^n\omega &= \int_{Y_r\times\mathbb{T}\times\prod_{i=r+1}^m\mathbb{T}}\log\|g_rs_r\|\pi_r^*c_1(\overline{\mathcal{O}(1)})^n\omega\\ &= \int_{Y_r\times\prod_{i=r+1}^m\mathbb{T}}\left(\int_{\mathbb{T}}\log\|g_rs_r\|c_1(\overline{\mathcal{O}(1)})^n\right)\omega\\ &= \int_{Y_r\times\prod_{i=r+1}^m\mathbb{T}}\left(\int_{\mathbb{T}}\log\|g_rs_r\|d\sigma_0(x)\right)\omega.\end{aligned}$$

The first equality follows since a Zariski closed subset with empty interior is a nullset with respect to a measure associated to differential forms.

We claim that the inner integral $\int_{\mathbb{T}}\log\|g_rs_r\|d\sigma_0(x)$ vanishes at each fibre. Recall that $g_r(t,t_1,\dots,t_m)=f_r(tt_r^{-1})$. We compute:

$$\begin{aligned}\log\|s_r(t,t_1,\dots,t_m)\| &= \int_{\mathrm{trop}^{-1}(\mathrm{trop}(t,t_1,\dots,t_m))} -\log|g_r(x)|\,\mathrm{d}\sigma_{(t,t_1,\dots,t_m)}(x)\\ &= \int_{\mathrm{trop}^{-1}(\mathrm{trop}(t,t_r))\subseteq(\mathbb{T}\times\mathbb{T}_r)^{\mathrm{an}}} -\log|f_r(xx_r^{-1})|\,\mathrm{d}\sigma_{(t,t_r)}(x,x_r)\\ &= \int_{\mathrm{trop}^{-1}(\mathrm{trop}(tt_r^{-1}))\subseteq\mathbb{T}_r^{\mathrm{an}}} -\log|f_r(x)|\,\mathrm{d}\sigma_{tt_r^{-1}}(x) = \rho_r\circ\mathrm{trop}(tt_r^{-1}),\end{aligned}$$

where ρ_r denotes the Ronkin function for f_r. Consider the fibre over the element

$$(t,t_1,\dots,t_{r-1},t_{r+1},\dots,t_m)\in Y_r\times\prod_{i=r+1}^m\mathbb{T}.$$

The integral over this fibre is

$$\begin{aligned}&\int_{\mathbb{T}}\log\|g_r(t,t_1,\dots,t_m)s_r(t,t_1,\dots,t_m)\|d\sigma_0(t_r)\\ &\qquad= \int_{\mathbb{T}}\left(\log|f_r(tt_r^{-1})|+\rho_r\circ\mathrm{trop}(tt_r^{-1})\right)d\sigma_0(t_r)\\ &\qquad\qquad= -\rho_r\circ\mathrm{trop}(t)+\rho_r\circ\mathrm{trop}(t)=0.\end{aligned}$$

□

We have now expressed the limit height as an arithmetic intersection number. When the $\overline{D}_i$ are toric, this intersection can be evaluated by convex geometry. It remains to relate this to the exact quantity predicted by Gualdi's conjecture. We phrase our results in terms of divisors, which are better suited to the toric intersection formula, instead of line bundles. Note that it is permissible to pull back divisors along p and q because they are both surjective.

Lemma 4.5. *Let $\overline{R(f_i)}$ denote the Ronkin divisor of f_i on T and assume that $\overline{D}_0,\dots,\overline{D}_{n-m}$ are semipositive toric adelic divisors on T. Then*

$$\begin{aligned}&\widehat{\deg}\left(\overline{R(g_1)},\dots,\overline{R(g_m)},q^*\overline{D}_0,\dots,q^*\overline{D}_{n-m},p^*\overline{P},\dots,p^*\overline{P}\mid X\right)\\ &\qquad= c\cdot\widehat{\deg}(\overline{R(f_1)},\dots,\overline{R(f_m)},\overline{D}_0,\dots,\overline{D}_{n-m}\mid T).\end{aligned}$$

Proof. Write $M_X=M_T\oplus L$ for the character lattice of the dense torus of X, where $L:=\bigoplus_{i=1}^m M_i$ and each M_i is the character lattice of the i-th parameter-torus factor, and let $\pi\colon(M_X)_{\mathbb{R}}\to(M_T)_{\mathbb{R}}$ be the projection. Denote the polytopes associated to $\overline{D}_i$ by $\Delta_i\subset(M_T)_{\mathbb{R}}$ and the associated v-adic roof functions by $\theta_{i,v}$. We denote

the polytope associated to $\overline{P} = \sum_i \pi_i^* \overline{\mathcal{O}(1)}$ by $\Delta_P \subset L_{\mathbb{R}}$. It is the product of m standard n-simplices. Indeed, the polytope of $\pi_i^* H_\infty$ is the standard n-simplex in $(M_i)_{\mathbb{R}}$, and the polytope of $P = \sum_i \pi_i^* H_\infty$ is their Minkowski sum; since they lie in complementary summands of $L_{\mathbb{R}} = \bigoplus_i (M_i)_{\mathbb{R}}$, this Minkowski sum is their Cartesian product. Because the metric is canonical the associated roof function is precisely 0_{Δ_P}. Viewing M_T and L as subspaces of M_X, the polytopes and roof functions of $q^*\overline{D}_i$ and $p^*\overline{P}$ agree with the ones of $\overline{D}_i$ and $\overline{P}$.

By Theorem 3.6 applied to X, the left hand side of the equality in question is

$$\sum_{v \in \Omega_K} n_v \operatorname{MI}_{M_X}\big(\rho_{g_1,v}^\vee, \dots, \rho_{g_m,v}^\vee, \theta_{0,v}, \dots, \theta_{n-m,v}, \underbrace{0_{\Delta_P}, \dots, 0_{\Delta_P}}_{mn}\big).$$

We prove the identity place by place. Fix a place $v \in \Omega_K$. Since $mn = \operatorname{rk} L$ of the roof functions are identically zero and supported on $\Delta_P \subseteq L_{\mathbb{R}}$ we may apply Lemma 3.12 to obtain

$$\begin{aligned}&\operatorname{MI}_{M_X}\big(\rho_{g_1,v}^\vee, \dots, \rho_{g_m,v}^\vee, \theta_{0,v}, \dots, \theta_{n-m,v}, \underbrace{0_{\Delta_P}, \dots, 0_{\Delta_P}}_{mn}\big)\\ &\qquad = \operatorname{MV}_L(\Delta_P, \dots, \Delta_P) \cdot \operatorname{MI}_{M_T}\big(\pi_*\rho_{g_1,v}^\vee, \dots, \pi_*\rho_{g_m,v}^\vee, \pi_*\theta_{0,v}, \dots, \pi_*\theta_{n-m,v}\big).\end{aligned}$$

The first factor is computed as

$$\operatorname{MV}_L(\Delta_P, \dots, \Delta_P) = (mn)!\, \operatorname{vol}_L(\Delta_P) = (mn)!/(n!)^m = \deg_P((\mathbb{P}^n)^m) = c.$$

Moreover $g_i = K[\phi_i](f_i)$ for the lattice embedding $\phi_i \colon M_T \hookrightarrow M_X$ given by

$$a \mapsto (a, 0, \dots, 0, -a, 0, \dots, 0),$$

with $-a$ in the i-th block, so $\pi \circ \phi_i = \mathrm{id}_{M_T}$. By Lemma 3.10, $\pi_*\rho_{g_i,v}^\vee$ is $\rho_{f_i,v}^\vee$, that is the roof function of $\overline{R(f_i)}$. Likewise $\pi_*\theta_{j,v} = \theta_{j,v}$. Reassembling via Theorem 3.6 on T, we obtain

$$\begin{aligned}&\widehat{\deg}\Big(\overline{R(g_1)}, \dots, \overline{R(g_m)}, q^*\overline{D}_0, \dots, q^*\overline{D}_{n-m}, p^*\overline{P}, \dots, p^*\overline{P} \mid X\Big)\\ &\qquad = \sum_{v \in \Omega_K} n_v \operatorname{MI}_{M_X}\big(\rho_{g_1,v}^\vee, \dots, \rho_{g_m,v}^\vee, \theta_{0,v}, \dots, \theta_{n-m,v}, \underbrace{0_{\Delta_P}, \dots, 0_{\Delta_P}}_{mn}\big)\\ &\qquad = c \cdot \sum_{v \in \Omega_K} n_v \operatorname{MI}_{M_T}\big(\rho_{f_1,v}^\vee, \dots, \rho_{f_m,v}^\vee, \theta_{0,v}, \dots, \theta_{n-m,v}\big)\\ &\qquad = c \cdot \widehat{\deg}(\overline{R(f_1)}, \dots, \overline{R(f_m)}, \overline{D}_0, \dots, \overline{D}_{n-m} \mid T).\end{aligned}$$

□

By combining the above results, we may conclude the proof of Gualdi's conjecture.

Theorem 4.6. *Let $1 \leqslant m \leqslant n$, let $f_1, \dots, f_m \in K[M]$ be nonzero Laurent polynomials, and let T be a projective toric variety with dense torus $\mathbb{T} = \mathbb{G}_m^n$. For each i, let*

$$V_i := V(f_i)$$

be the closure in T of the hypersurface defined by f_i. Let $u_j = (u_{1,j}, \dots, u_{m,j}) \in \mathbb{T}^m(\overline{\mathbb{Q}})$ be a generic sequence which is small for the standard Weil height on $\mathbb{T}^m$, and let $\overline{D}_0, \dots, \overline{D}_{n-m}$ be semipositive toric adelic divisors on T. Write $\theta_{k,v}$ for the v-adic roof function of $\overline{D}_k$, and $\rho_{i,v}^\vee$ for the v-adic Ronkin roof function of f_i. For all sufficiently large j, the translated hypersurfaces

$$u_{1,j}V_1, \dots, u_{m,j}V_m$$

meet properly. Denote their intersection cycle by $Z_j := (u_{1,j}V_1)\cdots(u_{m,j}V_m)\cdot\mathbb{T}_{\overline{\mathbb{Q}}}$. *Then*

$$\lim_{j\to\infty}\widehat{\deg}\left(\overline{D}_0,\dots,\overline{D}_{n-m}\mid Z_j\right)=\widehat{\deg}\left(\overline{D}_0,\dots,\overline{D}_{n-m},\overline{R(f_1)},\dots,\overline{R(f_m)}\mid T\right)$$
$$=\sum_{v\in\Omega_K}n_v\ \mathrm{MI}_M\left(\theta_{0,v},\dots,\theta_{n-m,v},\rho_{1,v}^\vee,\dots,\rho_{m,v}^\vee\right).$$

Here the middle intersection number may be computed on any common projective toric refinement of T *on which the Newton polytopes* $\mathrm{NP}(f_i)$ *define Cartier divisors; by the projection formula, its value is independent of this choice.*

Proof. Both sides of the identity are seen to vanish if one of the f_i is invertible. We may therefore assume all the f_i to be non-invertible. Choose a common projective refinement of the fan of T and the normal fans of $\mathrm{NP}(f_1),\dots,\mathrm{NP}(f_m)$; by the projection formula, we may replace T by the resulting toric variety and thus assume that every $\mathrm{NP}(f_i)$ defines a Cartier divisor on T. Lemma 4.2 identifies the limit with c^{-1} times an arithmetic intersection number on the universal family $\mathcal{V}_U\to U$, which may be computed on its closure $\mathcal{V}\subset X$. Lemma 4.4 replaces this intersection by the corresponding intersection on X involving the Ronkin divisors $\overline{R(g_1)},\dots,\overline{R(g_m)}$, and Lemma 4.5 identifies the latter with

$$c\,\widehat{\deg}\left(\overline{R(f_1)},\dots,\overline{R(f_m)},\overline{D}_0,\dots,\overline{D}_{n-m}\mid T\right).$$

The factors c^{-1} and c cancel, and Theorem 3.6 states that

$$\widehat{\deg}\left(\overline{D}_0,\dots,\overline{D}_{n-m},\overline{R(f_1)},\dots,\overline{R(f_m)}\mid T\right)=\sum_{v\in\Omega_K}n_v\ \mathrm{MI}_M\left(\theta_{0,v},\dots,\theta_{n-m,v},\rho_{1,v}^\vee,\dots,\rho_{m,v}^\vee\right),$$

so the asserted formula follows. □

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